

Total No. of Questions : 8]

SEAT No. :

**PB2306**

**[6263]-145**

[Total No. of Pages :2

**B.E. (Electronics/ E & TC)**

**REMOTE SENSING**

**(2019 Pattern) (Semester-VIII) (Elective - VI) (404192 C)**

*Time : 2½ Hours]*

*[Max. Marks : 70*

*Instructions to the candidates:*

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagram must be drawn wherever necessary.
- 3) Figures to the right indicates full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) With suitable diagram and example, explain various types of satellite data products. [8]
- b) Elaborate satellite data processing products offered by space agencies.[8]

OR

- Q2)** a) Explain Geometric corrections and atmospheric corrections and how they are removed during preprocessing. [8]
- b) With suitable illustrations, explain elements of visual interpretation used in remote sensing. [8]
- Q3)** a) With reference to any one real life application, elaborate suitability of microwave remote sensing. [8]
- b) Elaborate details of SAR and SLAR. Explain characteristic and interpretation of SAR data. [10]

OR

- Q4)** a) Explain principles of LIDAR operation. Give list of applications along with suitability of types. [10]
- b) Explain concept of microwave polarization in synthetic Aperture RADAR. [8]

*P.T.O.*

- Q5)** a) Give name of India's own navigation system and its key features, application areas with suitable diagram of satellite positioning. [10]  
b) Explain vulnerabilities of GNSS and its signal structure. [8]

OR

- Q6)** a) Explain Ground based augmentation system and space based augmentation system. [10]  
b) Explain navigation message of GNSS signal with suitable diagram. [8]

- Q7)** Elaborate application of Remote sensing with suitable use case and methodology. [18]

- a) Environmental Management  
b) Forestry

OR

- Q8)** Elaborate application of Remote sensing with suitable use case and methodology. [18]

- a) Urban Sprawl  
b) Flooding disaster

